

Research Idea: Candidate_Black_Hole_X-ray_Binary

ALMA Thesis Planner (automated pipeline)

A closure-validated systematics floor for sub-mm fast timing of an outbursting black hole X-ray binary: fractional rms, PSD slope, and a red-noise-referenced event test on ALMA Band 6 ToO data (2021.1.01208.T)

Motivation and positioning. A BHXB’s compact jet radiates in the sub-mm from the region where the outflow first becomes optically thin, so the amplitude and characteristic timescale of sub-mm flickering constrain the size, replenishment rate, and energetics of that zone. Sub-mm fast timing of outbursting BHXBs is an established niche with published fractional-rms and flare results, including from this proposal’s own group; this thesis is not a first measurement and is not framed as one. Its contribution is a *rigorously floor-calibrated* measurement on a new source and at new outburst stages, with the instrumental systematics treated as a first-class result rather than an assumed-negligible background. The parent proposal’s headline science is a radio-through-X-ray joint-fitting program — multi-person, multi-year. What one student can own end-to-end is the single-band question: *how strongly does the sub-mm jet vary above a defensibly derived instrumental floor, on what timescales, and is there anything in the light curves that a stationary red-noise process cannot produce?* That answer requires no other telescope, no new observations, and is a prerequisite for every downstream multiwavelength claim.

Step 0 — Weeks 1–3: archive retrieval, calibration restoration, and audit, before the analysis plan is locked. The archive delivers raw data plus calibration products, not ready-to-run calibrated MSs. Restoration requires the matching CASA version, and a ToO with possible manual calibration is version-brittle; two weeks are budgeted explicitly for this and success is *confirmed* before anything downstream is committed to. Then count the execution blocks actually delivered (ToO programs are routinely partly executed) and tabulate per epoch: native integration time (the MS cannot be rebinned finer than what was written), scan pattern and target-scan length, **whether a check source was observed**, spectral setup, flagging fraction, and mean target flux density. Three branch points are declared and each has a written plan: EB count (≥ 5 / 3–4 / 1–2), check-source presence, and target brightness (tens-of-mJy vs few-mJy). **The expectations branch with them.** At the end of week three the student and advisor sign off on which outcome is in play, including the explicit possibility that the guaranteed deliverable is a systematics floor plus upper limits — a methods thesis, agreed in month one rather than discovered in month eight. The audit table is deliverable (i).

Step 1 — Visibility-domain light curves. Fit a point source directly to the target visibilities in successive time bins at the native integration time and at logarithmically spaced rebinnings up to the scan length — photon-efficient and free of deconvolution artefacts. Self-calibration, if used at all, is phase-only on solution intervals *longer* than the timescales under study; amplitude self-calibration is forbidden and the prohibition is enforced in the extraction script, since it would absorb the signal into the gain table.

Step 2 — A non-circular systematic floor (core of the thesis). The phase calibrator cannot simply be pushed through the same pipeline: it is the source the gains are derived from, so its corrected light curve is flat nearly by construction and would license marginal detections. Three independent estimators are used instead. (a) **Closure amplitudes on the target — the primary internal estimator.** Closure quantities are immune to antenna-based gain errors by construction, including atmospheric phase/amplitude decorrelation, and are computable from the same MSs with no extra machinery. Closure-amplitude time series therefore isolate genuine changes in the source’s visibility structure from gain-driven artefacts, and set the internal floor directly. (b) **Gain cross-validation, always.** Solve gains on alternating subsets of calibrator scans, apply them to the held-out scans, and use the scatter in held-out corrected fluxes to measure the real transfer error as a function of scan separation. (c) **Check source, if Step 0 finds one** — the clean external case, processed identically to the target. Antenna-drop and split-half baseline jackknives are retained only as a cross-check, not as evidence: in compact configurations neighbouring antennas share much of the same troposphere, so their “non-common-mode” assumption partly fails. On hour-long timescales, where interleaved sampling gives little leverage, supplement with regression of residual flux and gain statistics against elevation, airmass, PWV, and flagging fraction. Output: a fractional-rms systematic floor *as a function of timescale*, above which — and only above which — target structure is claimed astrophysical.

Step 3 — Ensemble timing statistics, the primary science result. Above the floor: fractional rms versus timescale; the power spectral density fitted **slope-only**, over an accessible frequency range quoted explicitly from the Step 0 audit (roughly native integration time to scan length, ~ 1.5 decades), with thermal noise subtracted. No break fitting — the dynamic range does not support it. Because phase-referenced scans impose hard periodic gaps whose window function aliases power, the method is validated by generating red-noise light curves of known slope, passing them through the *actual* sampling pattern, flagging, and noise level, and confirming slope recovery; Lomb–Scargle plus this forward-modelling validation replaces any naive periodogram. Test also for a linear rms–flux relation, which discriminates multiplicative propagating-fluctuation variability inherited from the accretion flow against additive discrete-ejection variability. At the few-mJy end this yields upper limits, per the Step 0 agreement.

Step 4 — A non-redundant event test, framed correctly. If the variability is red noise, “flare rate” and “fraction of flux in discrete events” are threshold-dependent restatements of the Step 3 slope and normalisation, not independent physical quantities. So the census is posed as a falsification test of the Step 3 model, not as a parallel result. Using the large ensemble of simulated red-noise realisations already built in Step 3 — passed through the real sampling and the Step 2 floor — the student asks two specific questions: (i) is there an **excess of high-significance excursions** in the data over the simulated distribution at matched threshold, and (ii) do individual events show **rise/decay asymmetry** beyond what the stationary stochastic realisations produce? Only excursions that fail the simulated null are reported as discrete ejection candidates, and for those the amplitude, rise and decay times, and fluence are measured as physical constraints on the emitting region. If nothing exceeds the null, the honest statement is that the light curves are consistent with a single stochastic process of the measured slope — itself a clean result, and reported as such. Sideband-relative fluxes are checked once for consistency with unity at the precision the floor analysis independently establishes and reported in one sentence; no spectral-index-versus-time science is attempted, as the $\sim 7\%$ lever arm cannot support it.

Step 5 — Evolution across the outburst. Repeat Steps 2–4 for every usable epoch and plot systematic floor, fractional rms (or limit), PSD slope, and any null-failing event properties against

mean sub-mm flux, which orders epochs by outburst stage. This delivers the proposal’s ”as a function of accretion rate” ambition in the one band the student controls. With 3–4 usable EBs it reduces to a two-point comparison; with 1–2, Step 5 is dropped and the thesis becomes a single-epoch deep study — full restoration, floor derivation, rms/PSD-slope measurement, and event test on the best block — a branch the Step 0 audit fires in month one.

Deliverables. (i) The restoration log and EB/scan-structure audit table; (ii) a reusable visibility-domain extraction pipeline with closure-amplitude and gain-cross-validation floor estimators and the red-noise-through-real-sampling simulator; (iii) a per-epoch table of floor, fractional rms or limit, PSD slope, and event-test outcomes with honest uncertainties; (iv) figures of these versus outburst stage.

Scope notes. The target is inherited from whatever the ToO trigger fired on; the plan is deliberately target-agnostic. The simultaneous radio/IR/optical/far-UV/X-ray data mentioned in the abstract are not part of the described dataset and are assumed nowhere; cross-band lag work is future work for the collaboration. Everything runs in CASA plus standard Python.